Research Highlights

- Four packet-level attacker actions are modeled as Lie-cone generators on flows.
- Trajectories in the cone provably preserve rate, length, header and MTU invariants.
- On full CSE-CIC-IDS2018/2017 and UNSW-NB15, FFE evades 100/96/62% of attacks.
- Feature-space baselines such as HopSkipJump yield 0% physically feasible flows.
- Lie-guided adversarial training cuts FFE success from 96.3% to 33.0%.